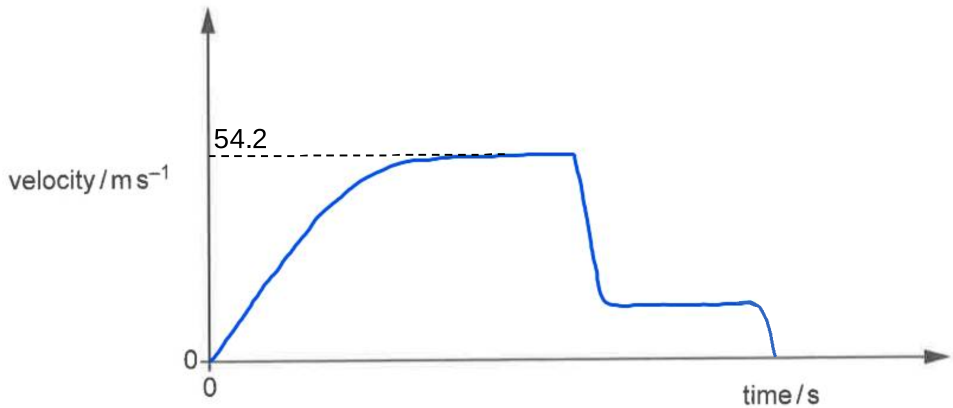
	$v_E = 2.28\ v_J$
5(a)(i)	The velocity is obtained from the gradient of the tangent of the graph at 0.60 s. $\text{gradient} = \text{velocity} = \dfrac{5.3-0}{1.2-0.30}$ $= 4.4\ \text{m s}^{-1}$

No.	Solution
5(a)(ii)	$\text{average velocity} = \dfrac{\text{total displacement}}{\text{total time}}$ $= \dfrac{4.9}{1.0} = 4.9 \text{ m s}^{-1}$
5(b)	At the start of the jump, the vertical velocity of the skydiver is zero and he only experiences acceleration of free fall due to weight. His speed increases as he falls and the upward air resistance acting on him increases with speed as well. The resultant force and his acceleration decreases. The air resistance increases until it is equal to the weight of the skydiver. At this point, the resultant force acting on him is zero and the acceleration drops to zero as well. He has reached terminal velocity and will continue to fall with this constant speed before the parachute opens.
5(c)	When skydiver falls with constant velocity, $\text{weight} = \text{air resistance}$ $(90)(9.81) = \dfrac{1}{2}(1.2)(0.50)v^2$ $v = 54.2 \text{ m s}^{-1}$
5(d)	The student is correct in stating that the parachute causes a large upward air resistance on the skydiver. However, the skydiver does not travel upwards. He is still moving downwards but his velocity decreases sharply due to a large deceleration caused by the upward force.
5(e)	
5(f)	Y will experience the same air resistance as X at the same speed. However Y has a larger weight hence he will reach a larger constant velocity and also take a longer time to reach it.
6(a)(i)	Electric field strength at a point is defined as the electric force exerted